

Puneet Velidi

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EDUCATION

University of Victoria

MSc in Statistics (thesis-based, fully-funded)

Co-supervisors: Prof. Farouk Nathoo & Prof. Michelle Miranda

Thesis: *Methods for Censored Covariates, Spatial Transcriptomic Data and Neural Manifolds*

Victoria, BC

Jan. 2025 – (anticipated) Dec. 2026

Cornell University

BA in Statistics & Economics

Ithaca, NY

Sep. 2018 – Dec. 2022

PUBLICATIONS ([†]indicates joint first authorship)

Puneet Velidi, Maryclare Griffin, Juxin Liu, Brian Richardson, Jing Qian, Roland Matsouaka, Sujit Ghosh, Tanya Garcia, Farouk Nathoo. *SPARTACCUS: Semiparametric Bayesian Proportional Hazards Model with a Censored Covariate in Huntington's Disease*. **In Preparation**, 2026.

Puneet Velidi, Zhengxiao Wei, Farouk Nathoo. *Covariance Nonstationarity is Evident in Spatial Transcriptomics and Provides a New Categorization of Spatially Varying Genes*. **Submitted for publication**, 2026. [bioRxiv](#)

Puneet Velidi, Michelle Miranda, Farouk Nathoo, Ashery Mbilinyi, Cédric Beaulac. *Shape Analysis as an Inductive Bias for Glioma Grading from Tumor Contours*. **Submitted for publication**, 2026. [arXiv](#)

Puneet Velidi, Enrico Amico, Farouk Nathoo, Michelle Miranda. *Neural manifold connectomics reveals multiregime functional connectivity*. **Under review**, 2026. [bioRxiv](#)

Fabio Luchese, **Puneet Velidi**, . . . , Paolo Cassano. *Transcranial Photobiomodulation on Language and Cognitive Performance in Down Syndrome: A Pilot Randomized Sham-Controlled Trial*. **Under review**, 2026. [medRxiv](#)

B. Ferreras, . . . , **P. Velidi**, et al., P. Cassano. *Transcranial Photobiomodulation in Bipolar Disorder: A Case Series on Executive Skills and fMRI BOLD Signal*. **Submitted for publication**, 2026. Available upon request.

Guillermo Gonzalez-Garibay, **Puneet Velidi**, Akila Weerasekera, Nikos Makris, Carlos Lohmann, Katherine Anne Collins, Naomi Gaggi, Dan Vlad Iosifescu, Paolo Cassano. *Volumetric Changes Induced by Transcranial Photobiomodulation Therapy in Major Depressive Disorder: An Exploratory Neuroimaging Study*. **Submitted for publication**, 2026. Available upon request.

Puneet Velidi[†], Zhengxiao Wei[†], Shreena N. Kalaria, Yimeng Liu, Céline M. Laumont, Brad H. Nelson, Farouk S. Nathoo. *Generalized Jeffreys's Approximate Objective Bayes Factor: Model Selection Consistency, Finite Sample Accuracy, and the Statistical Evidence in 71,126 Clinical Trial Findings*. **Under revision**, 2025. [arXiv](#)

S. Parker Singleton[†], **Puneet Velidi**[†], Louisa Schilling, Andrea I. Luppi, Keith Jamison, Linden Parkes, Amy Kuceyeski. *Altered structural connectivity and functional brain dynamics in individuals with heavy alcohol use elucidated via network control theory*. **Biological Psychiatry: Cognitive Neuroscience and Neuroimaging**, 2024. [Link](#)

SCHOLARSHIPS, AWARDS, RECOGNITIONS

2026	ICSA-Canada Chapter — Student/Postdoc Travel Scholarship Award This award supports travel expenses for the International Chinese Statistical Association – Canada Chapter 2026 Symposium in Montreal.	600 CAD
2025	Labs4 — Technology Readiness Level-Up (TRL [^]) Fellowship Lab-to-market grant; ChimeraML, combinatorial nanobody design with an ensemble of protein ML models for immune checkpoint inhibitor therapy. Funded by Mitacs, New Ventures BC, and BCIT. Labs4 TRL [^] Pitch Competition, BC Region, 2nd place.	15,000 CAD
2025	University of Victoria — Betty & Gilbert Kennedy Graduate Scholarship in Math. & Stats. Awarded to academically outstanding students in the Department of Mathematics and Statistics upon the recommendation by the department (<i>Donor Award</i>).	5,130 CAD
2025	Maud Menten Institute/PIMS — Research Accelerator Award This award supports a collaboration with the MGH Photobiomodulation Lab on Bayesian analysis of the TransPhoM-DS clinical trial.	5,000 CAD

2025	University of Victoria — FGS Graduate Fellowship (Master's) This fellowship is awarded to students of high academic standing upon recommendation by their department. <i>The typical M&S FGS Master's Fellowship is 2,500 CAD.</i>	<i>5,000 CAD</i>
2025	University of Victoria — FGS Travel Award & CUPE Conference Award These awards supported travel expenses for CNS 2025.	<i>1,050 CAD</i>
2025	3MC–PIMS–IDMS–ICMS — Summer School Supported Participant This recognition supported travel expenses + room and board for the two-week 2025 3MC–PIMS–IDMS–ICMS Summer School.	—
2025	Maud Menten Institute/PIMS — MMI HQP Summit Supported Participant This recognition supported travel expenses + room and board for the 2025 MMI HQP Summit.	—

COMPETITIONS

2026	Y Combinator (P26) Selected for an in-person interview in San Francisco out of 80k applications (approx. top 2.5%).	—
2026	Critical Ops: A DC National Security Hackathon (spon. OpenAI, Meta, Squadra Ventures) 1st place out of 166 participants , ENEL (drone swarm 3D Gaussian Splat). Won demo opportunities with OpenAI and Meta .	—
2026	SCSP AI+ Expo Hackathon 2nd place out of 200+ participants , 1st place DC region, Electrical Grid Track. Franklin, IoT grid sound anomaly detection & agent-to-agent load negotiation.	<i>5,000 USD</i>
2026	Stanford University RNA 3D Folding Competition Placed in the top 3% of 1,867 teams; Silver Medal .	—

PRESENTATIONS

Oral Presentations

Brain Fingerprinting of Principal-component derived fMRI BOLD functional connectivity. The 7th ICSA-Canada Chapter Symposium, August **2026**. Montreal, QC. Invited Talk, **Invited Session Chair**.

A Bayesian Semiparametric Model for Survival Analysis with a Right Censored Covariate. 9th International Conference on Econometrics and Statistics, August **2026**. Kyoto, JP. Co-author of invited talk.

A Bayesian Semiparametric Model for Survival Analysis with a Right Censored Covariate. 2026 Statistical Society of Canada Annual Meeting, May **2026**. Hamilton, ON. Contributed Talk.

“Applications of Protein Folding Models in Immunotherapeutic Research.” UBC Math Bio Seminar, January **2026**. Invited Seminar.

“70,000 Leagues Under the Sea of p -Values.” Maud Menten Institute / Mathematical and Statistical Biology Seminar, October **2025**. Invited Seminar.

“70,000 Leagues Under the Sea of p -Values.” UBC Stats Seminar, October **2025**. Invited Seminar.

Statistical methods for censored covariates. 2025 WNAR/IMS Annual Meeting, June **2025**. Whistler, BC. Co-author of discussant remarks for invited session.

“Network Control Theory in Neuroscience: Characterizing the Brain’s Energy Landscape.” Cognition and Brain Sciences Seminar, University of Victoria, March **2025**. Invited Seminar.

“Network Control Theory in Neuroscience: Characterizing the Brain’s Energy Landscape.” Maud Menten Institute / Mathematical and Statistical Biology Seminar, University of Victoria, February **2025**.

Poster Presentations

A Bayesian Semiparametric Model for Survival Analysis with a Right Censored Covariate. The 7th ICSA-Canada Chapter Symposium, August **2026**. Montreal, QC.

Bayesian Analysis of the TransPhoM-DS Clinical Trial. Organization for Human Brain Mapping 2026 Annual Meeting, June **2026**. Bordeaux, FR.

Ahab’s Harpoon: Calibrated irreproducibility detection with approximate objective Bayes factors. 2026 Statistical Society of Canada Annual Meeting, May **2026**. Hamilton, ON.

Reassessing the Statistical Evidence in Clinical Trials with Extended Approximate Objective Bayes Factors. Maud Menten Institute 2025 HQP Summit and Working Group Meetings, July **2025**. Edmonton, AB.

Photobiomodulation therapy-induced volumetric brain changes on patients with depression. Cognitive Neuroscience Society 2025 Annual Meeting, March **2025**. Boston, MA.

Altered structural connectivity and functional brain dynamics in individuals with heavy alcohol use elucidated via network control theory. Organization for Human Brain Mapping 2024 Annual Meeting, June **2024**. Seoul, SK.

Network control analysis of individuals with substance use disorder. Organization for Human Brain Mapping 2023 Annual Meeting, July **2023**. Montreal, QC.

RESEARCH EXPERIENCE

Graduate Researcher

Victoria, BC

University of Victoria — Department of Mathematics and Statistics

Jan. 2025 – Present

Nathoo Lab, PI: Prof. Farouk Nathoo & Prof. Michelle Miranda

Visiting Researcher

Birmingham, UK

University of Birmingham — School of Mathematics

Aug. 2025

AMICO Lab, PI: Prof. Enrico Amico

Postbacc Researcher

Boston, MA

Harvard Medical School & MGH — Division of Neuropsychiatry and Neuromodulation

May 2024 – Dec. 2024

MGH Photobiomodulation Laboratory, PI: Prof. Paolo Cassano

- Statistical analysis of TransPhoM-DS, TRIADE, TPEB, TRAP-AD clinical trials.

Undergraduate Researcher

Ithaca, NY

Weill Cornell Medicine — Department of Radiology

Jan. 2022 – Dec. 2022

Computational Connectomics Laboratory, PI: Prof. Amy Kuceyeski

PROFESSIONAL EXPERIENCE

Founder

Vancouver, BC & Washington, DC

ChimeraML — TRL Level-Up (Labs4) Fall 2025 Cohort

Sept. 2025 – Present

- Combinatorial nanobody design using an ensemble of protein ML models for immune checkpoint inhibitor therapy.
White paper available upon request.

Software Engineer II

Bentonville, AR

Walmart Global Tech — EBS

Feb. 2023 – May 2024

- *Early Career Recognition*, awarded to <10% of 2023 Walmart Global Tech Development Program.

AI/ML & NextGen Analytics Intern

New York, NY

Memorial Sloan Kettering Cancer Center — DigITs

June 2022 – Aug. 2022

TEACHING EXPERIENCE

Teaching Assistant

Victoria, BC

University of Victoria — Department of Mathematics and Statistics

Jan. 2025 – April 2025

Math Support Center Tutor

Ithaca, NY

Cornell University — Department of Mathematics

Sep. 2021 – Dec. 2022

WORKSHOPS AND SHORT COURSES

BIRS Workshop: A Roadmap towards Developing Mechanobiochemical Models for Single and Collective Cell Migration through Complex Non-Isotropic Environments. *Banff International Research Station (BIRS), Banff, AB.* May 10–15, **2026**.

Five-day invited workshop on 2- and 3-D mechanobiochemical models of single and collective cell migration.

eScience JASP Hackathon. *Netherlands eScience Center, Amsterdam, NL.* February 16–17, **2026**.

Developed a JASP module alongside the core JASP programming team.

3MC-PIMS-IDMS-ICMS Summer School on Quantitative Molecular and Cellular Biology. *University of Manitoba, Winnipeg, MB. June 16–27, 2025.*

Two-week school on quantitative methods for structural, molecular, and cellular biology, using cancer as an integrative example.

TECHNICAL REPORTS

Computational Design of Cross-Reactive Nanobodies Targeting the Human Chemokine Receptor Family. ChimeraML, **2026**. Available upon request.

Immunodynamic Modeling in Melanoma and Gastric Cancer. Supervised by Kang-Ling Liao and Michael Chan. 3MC-PIMS-IDMS-ICMS Summer School on Quantitative Molecular and Cellular Biology, **2025**.

PROFESSIONAL SERVICE

Network Neuroscience, Reviewer

Human Brain Mapping, Reviewer (with Supervisor)

JASA Applications and Case Studies, Reviewer (with Supervisor)